

Notes: Brown & Goolsbee (JPE, 2002)

Does the Internet Make Markets More Competitive? Evidence from the Life Insurance Industry

Contents

1	Big picture and incremental contribution	3
1.1	Incremental contribution of the paper	3
2	Data and measurement	3
2.1	Life-insurance setting	3
2.2	LIMRA policy-level transaction data	4
2.3	Price variable	4
2.4	Forrester Technographics Internet data	5
2.5	Price dispersion measure	5
3	Theory and empirical specifications	5
3.1	Search theory motivation	5
3.2	Baseline hedonic price regression	6
3.3	Internet-use price regression	6
3.4	Placebo and robustness specifications	6
3.5	Price-dispersion specification	6
4	Identification strategy	7
4.1	Core identification logic	7
4.2	Why term life is close to the ideal setting	7
4.3	Threats and responses	7
5	Tables and figures	8
5.1	Figure 1: Sample page from QuickQuote.com	8
5.2	Table 1: Summary statistics, 1992–97	8
5.3	Table 2: Basic specification	9
5.4	Table 3: Results by category: term prices	9
5.5	Table 4: Log real price as a function of Internet usage	10
5.6	Table 5: Further controls	10
5.7	Table 6: Price dispersion and Figure 2: Predicted standard deviation of price residuals	11
6	Interpretive synthesis	12
7	Short summary	12
8	Conclusion	12
8.1	Core conclusion	12

8.2	Economic intuition	12
8.3	Incremental contribution in one sentence	13
8.4	Limitations	13
8.5	Final takeaway	13

1 Big picture and incremental contribution

- **Question.** Did Internet comparison-shopping sites make an offline financial market more competitive by reducing consumer search costs?
- **Setting.** The paper studies U.S. term life insurance from 1992–1997, a market with high search costs, substantial markups, and a relatively homogeneous product that is comparable across firms.
- **Treatment intuition.** Starting in the mid-1990s, websites such as QuickQuote.com and Quotesmith.com made it much easier to compare term life premiums across insurers. The sites focused mainly on term life insurance, not whole life insurance.
- **Empirical strategy.** The authors merge micro data on individual life-insurance purchases from LIMRA with demographic-cell measures of Internet use and online insurance research from Forrester Technographics.
- **Main result.** Term life prices fell more for demographic groups in which Internet use or online insurance research grew more quickly. The paper estimates that Internet growth from 1995 to 1997 reduced term life prices by about 8–15 percent.
- **Placebo logic.** Prices did not fall with Internet use before the comparison sites existed, and prices of whole life policies, which were not covered by the comparison sites, did not show the same effect.
- **Price dispersion result.** Consistent with search theory, initial Internet adoption increased within-group price dispersion, but as online research became more common, dispersion fell.
- **Welfare calculation.** The authors estimate annual consumer-surplus gains of about \$115–\$215 million for new term life policies, with larger upper-bound gains if renewals are included.

1.1 Incremental contribution of the paper

- **First offline-price evidence for Internet competition.** Earlier Internet-commerce studies mostly compared online and offline prices or studied online price dispersion. This paper asks whether Internet search changed prices in an offline market.
- **Micro transaction-price design.** The paper uses individual policy transactions rather than advertised list prices, enabling hedonic controls for policy and buyer characteristics.
- **Search theory plus real market evidence.** It connects the theoretical prediction that lower search costs reduce markups to a large real-world financial product market.
- **Product-market placebo innovation.** The contrast between term life and whole life insurance provides a built-in falsification test: the Internet comparison sites covered the former but not the latter.
- **Timing placebo innovation.** Internet use before the comparison websites existed should not affect prices. The paper uses this timing to distinguish general Internet diffusion from comparison-shopper competition.
- **Dispersion contribution.** The paper studies not only average prices but also within-group price dispersion, showing a non-monotonic pattern predicted by search models when the initially informed share is close to zero.
- **Financial-market relevance.** It documents that search-cost reductions can matter in retail financial markets, not only in conventional consumer goods markets such as books or cars.

2 Data and measurement

2.1 Life-insurance setting

The paper focuses on individual life-insurance policies. The key product distinction is between term life insurance and whole life insurance.

- **Term life.** Provides insurance coverage for a specified period, such as one or five years. It has no cash value after the term ends and is relatively homogeneous across firms.
- **Whole life.** Provides lifetime coverage and includes a savings component. It is more differentiated and was not covered by the main Internet comparison sites during the sample period.
- **Why term life is useful.** Term policies are easier to compare across insurers, so Internet comparison sites could sharply reduce search costs.

Interpretation: The design needs a product for which online comparison sites plausibly reduce search costs without simultaneously changing the product itself. Term life insurance fits this condition much better than whole life insurance.

2.2 LIMRA policy-level transaction data

LIMRA International conducts annual Buyer Studies of individual life-insurance purchases. The authors combine Buyer Studies from 1992–1997. The data include:

- premiums and face values;
- policy type, term length, rated status, participating status, and riders;
- policyholder demographics such as age, state, occupation, and income;
- detailed transaction-level observations of actual purchased policies.

The main term-life sample restricts to level term policies that:

- insure only one person;
- are owned by the premium payer;
- involve insured individuals age 20–75;
- have no additional riders;
- have terms of five years or less, to avoid regulatory confounds for longer-term guarantees.

Interpretation: These restrictions make the product as homogeneous as possible. They also reduce the risk that observed price changes reflect changes in coverage features rather than competition.

2.3 Price variable

The dependent variable in the main price regressions is

$$\log \left(\frac{\text{annual premium}}{\$1,000 \text{ of face value}} \right).$$

The authors control for many policy and individual characteristics, including amount, term length, marital status, face value, state, age, occupation, interest rates, and other policy attributes.

Interpretation: The hedonic price design compares quality-adjusted prices, not raw premiums. This is essential because insurance prices vary mechanically with age, health/rated status, smoker status, face amount, and term length.

2.4 Forrester Technographics Internet data

The authors use Forrester’s 1998 Technographics survey, which includes nearly 100,000 individuals and asks about computer ownership, Internet use, online buying, and product research. The survey also includes age, state, occupation, and income, which can be matched to LIMRA demographic cells.

The authors construct several group-level Internet measures:

- share online by age-state-year;
- share online by age-occupation-year;
- share online by occupation-state-year;
- share online by age-income-year;
- share that both has Internet access and has researched insurance online.

Internet use is measured retrospectively from Forrester. For 1992 and 1993, it is scaled using Internet host growth because early retrospective measures are limited. Since the comparison sites did not exist until 1996, the baseline Internet-treatment variable is set to zero before 1996 and then equals the group-level online share or insurance-research share after 1996.

Interpretation: This construction turns Internet diffusion into a demographic-cell exposure variable. Identification comes from the fact that different age-state-occupation-income groups adopted the Internet at different speeds.

2.5 Price dispersion measure

To study price dispersion, the authors first estimate a hedonic price regression and compute residual prices. For each age-state-year group, they compute the standard deviation of residual prices. They then regress this within-group dispersion measure on the share of group members researching insurance online, its square, and its cube.

Interpretation: This removes observable product and buyer differences before measuring dispersion. The dispersion outcome asks whether identical-looking buyers in the same demographic group face more or less transaction-price heterogeneity as online search expands.

3 Theory and empirical specifications

3.1 Search theory motivation

The paper relies on a standard search-cost logic. Suppose there are informed consumers who know all prices and uninformed consumers who must search at positive cost. If more consumers become informed, firms face stronger competitive pressure.

- If many consumers are uninformed, firms can sustain high markups.
- When a small group becomes informed, some firms may cut prices to attract them, while other firms continue exploiting uninformed consumers, increasing price dispersion.
- When the informed share becomes large enough, high-price firms lose too many customers, and average prices and price dispersion fall.

Interpretation: This explains why the paper predicts lower average prices but a non-monotonic effect on price dispersion. At the start of Internet adoption, the informed share is close to zero, so dispersion can rise before falling.

3.2 Baseline hedonic price regression

The basic regression can be summarized as:

$$\log(Premium_i/FaceValue_i) = X_i'\beta + \delta_t + \varepsilon_i,$$

where X_i contains policy and individual characteristics and δ_t are year effects.

Interpretation: Table 2 uses this specification to document the large decline in term life prices relative to 1992 and to compare term life with whole life.

3.3 Internet-use price regression

The central empirical specification is:

$$\log(Premium_i/FaceValue_i) = X_i'\beta + \delta_t + \theta InternetUse_{g(i),t} + \varepsilon_i,$$

where $g(i)$ is a demographic group such as age-state, age-occupation, age-income, or occupation-state.

In the preferred later specifications, the Internet variable is the share of people in a demographic group who have Internet access and report researching insurance online.

Interpretation: The coefficient θ measures whether prices fall faster for groups that are more likely to use the Internet for insurance search after comparison sites appear.

3.4 Placebo and robustness specifications

The paper tests several alternatives:

- using whole life policies instead of term life policies;
- measuring online use in the pre-comparison-site years 1992–1995;
- controlling for mortality changes by group;
- allowing nonlinear effects of Internet research;
- interacting Internet use with year dummies;
- testing whether Internet use predicts riders or other unmeasured policy quality.

Interpretation: These tests try to rule out secular price declines, unobserved mortality improvements, changes in policy composition, or general Internet adoption unrelated to comparison shopping.

3.5 Price-dispersion specification

Let $\hat{\varepsilon}_{g,t}$ be the residuals from the hedonic price regression within group g and year t . The paper estimates:

$$SD(\hat{\varepsilon}_{g,t}) = \alpha + \beta_1 Research_{g,t} + \beta_2 Research_{g,t}^2 + \beta_3 Research_{g,t}^3 + \eta_{g,t}.$$

Interpretation: The cubic form allows price dispersion to rise at very low informed shares and then decline after online search becomes widespread.

4 Identification strategy

4.1 Core identification logic

The paper uses differential Internet adoption across demographic groups and time. The key comparison is whether prices fall more rapidly for groups more likely to use the Internet after comparison-shopping websites become available.

- **Cross-group variation:** age, state, occupation, and income groups adopt the Internet at different rates.
- **Timing variation:** insurance comparison sites begin in the mid-1990s, especially after 1996.
- **Product variation:** term life is covered by comparison sites; whole life is not.
- **Outcome variation:** average price and dispersion should respond differently according to search theory.

Interpretation: This is a difference-in-differences style design: groups with larger Internet exposure after the relevant websites appear should have larger price declines, while pre-period Internet use and noncovered products should not show the same relationship.

4.2 Why term life is close to the ideal setting

- Term life is relatively standardized and price-comparable.
- The product is financially important and historically expensive to search.
- Comparison websites focused on term life quotes.
- Insurers generally filed rates with state regulators, making immediate one-to-one online price discrimination less likely during the sample period.

Interpretation: These features make price changes more plausibly attributable to search-cost reductions rather than product redesign or direct online discounting.

4.3 Threats and responses

- **Mortality improvements:** if mortality improved faster for Internet-adopting groups, prices could fall for actuarial reasons. Table 5 controls for mortality and still finds effects.
- **General price trends:** Table 2 and Table 3 show broad term price declines, but Table 4 links the declines to Internet-use variation.
- **Non-Internet secular trends:** pre-1996 Internet use does not predict price declines before comparison sites existed.
- **Policy quality changes:** Table 5 tests riders and whole life policies and finds that these do not explain the core results.
- **Unobserved insurer quality:** the paper cannot include firm fixed effects because LIMRA lacks company identifiers; this remains a limitation.

Interpretation: The identification is credible because the timing, product placebo, pre-period placebo, and dispersion pattern all point to search-cost reductions. The main limitation is that company identities and some policy-quality details are not observed.

5 Tables and figures

5.1 Figure 1: Sample page from QuickQuote.com

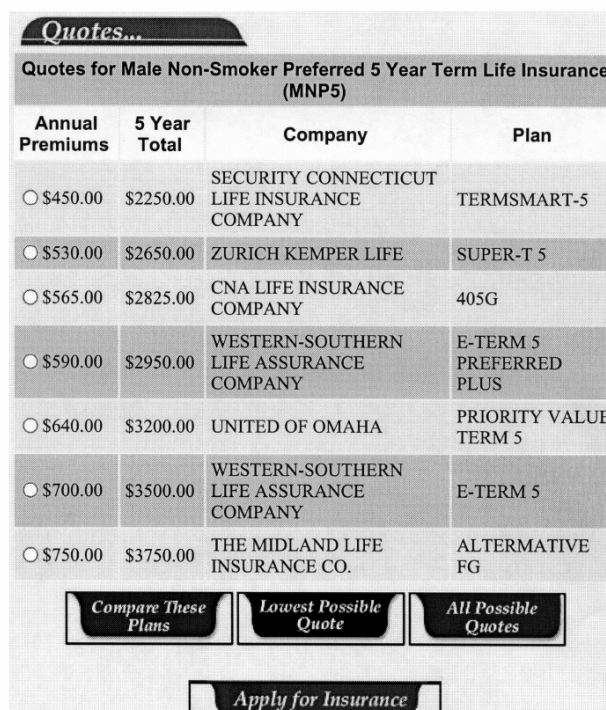
Read: The figure shows a term-life comparison page listing insurers, annual premiums, five-year totals, company names, and plans. It illustrates how consumers could compare many insurers at once.

Interpretation: The figure is not evidence by itself; it is a mechanism illustration. It shows the type of information shock that lowered search costs in the mid-1990s.

Economic message: A consumer who previously needed an agent or many phone calls could now see price rankings quickly, creating pressure on insurers with high premiums.

486

JOURNAL OF POLITICAL ECONOMY



Quotes...			
Quotes for Male Non-Smoker Preferred 5 Year Term Life Insurance (MNP5)			
Annual Premiums	5 Year Total	Company	Plan
○ \$450.00	\$2250.00	SECURITY CONNECTICUT LIFE INSURANCE COMPANY	TERMSMART-5
○ \$530.00	\$2650.00	ZURICH KEMPER LIFE	SUPER-T 5
○ \$565.00	\$2825.00	CNA LIFE INSURANCE COMPANY	405G
○ \$590.00	\$2950.00	WESTERN-SOUTHERN LIFE ASSURANCE COMPANY	E-TERM 5 PREFERRED PLUS
○ \$640.00	\$3200.00	UNITED OF OMAHA	PRIORITY VALUE TERM 5
○ \$700.00	\$3500.00	WESTERN-SOUTHERN LIFE ASSURANCE COMPANY	E-TERM 5
○ \$750.00	\$3750.00	THE MIDLAND LIFE INSURANCE CO.	ALTERNATIVE FG

FIG. 1.—Sample page from Quickquote.com

Figure 1: Sample page from QuickQuote.com.

5.2 Table 1: Summary statistics, 1992–97

Read: The term-life sample has 10,812 observations. The mean premium per \$1,000 of face value is 3.62, the mean policy amount is \$132,970, and the average policy length is 2.27 years. About 77 percent are nonsmokers, 67 percent are male, and 17 percent of the relevant group is online.

Interpretation: The sample is a policy-level transaction sample, not a web-scraped quote sample. That is what lets the paper study actual purchases and offline market prices.

Economic message: The summary table establishes that the paper studies real term life purchases with meaningful demographic and policy variation.

TABLE 1
SUMMARY STATISTICS, 1992–97

Type	Term
Premium/(\$1,000 of face)	3.62 (4.91)
Real amount of policy (in thousands of 1990 dollars)	132.97 (136.41)
Length of policy	2.27 (1.86)
Nonsmoker	.77 (.42)
Male	.67 (.47)
Policy is rated	.08 (.27)
R^{length}	.88 (.10)
Participating policy	.88 (.32)
% on-line	.17 (.14)
Observations	10,812

SOURCE.—Authors' calculations using data from LIMRA International and Forrester.

NOTE.—Standard errors are in parentheses.

Annual Premiums	5 Year Total	Company	Plan
☐ \$450.00	\$2250.00	SECURITY CONNECTICUT LIFE INSURANCE COMPANY	TERMSMART-5
☐ \$530.00	\$2650.00	ZURICH KEMPER LIFE	SUPER-T 5
☐ \$565.00	\$2825.00	CNA LIFE INSURANCE COMPANY	405G
☐ \$590.00	\$2950.00	WESTERN-SOUTHERN LIFE ASSURANCE COMPANY	E-TERM 5 PREFERRED PLUS
☐ \$640.00	\$3200.00	UNITED OF OMAHA	PRIORITY VALUE TERM 5
☐ \$700.00	\$3500.00	WESTERN-SOUTHERN LIFE ASSURANCE COMPANY	E-TERM 5
☐ \$750.00	\$3750.00	THE MIDLAND LIFE INSURANCE CO.	ALTERNATIVE FG

Compare These Plans Lowest Possible Quote All Possible Quotes

Apply for Insurance

FIG. 1.—Sample page from Quickquote.com

5.3 Table 2: Basic specification

Read: Column 1 shows term life prices fall sharply in 1996 and 1997 relative to 1992, with the 1997 term coefficient around -0.2702 . Column 2 shows whole life does not exhibit the same late-sample decline; the 1997 coefficient is approximately zero.

Interpretation: This is the first key placebo. Both term and whole life prices drift down before 1996, but only term life drops sharply after comparison sites become relevant.

Economic message: The contrast supports the interpretation that the Internet affected the product covered by comparison sites rather than all life-insurance products.

5.4 Table 3: Results by category: term prices

Read: Price declines are larger in high-Internet-adoption groups. Fast-adopting states show a much larger 1997 decline than slow-adopting states. High-skill occupations and younger buyers also show larger declines.

TABLE 2
BASIC SPECIFICATION

Type	Term (1)	Whole (2)
D93	.0609 (.0133)	-.0597 (.0134)
D94	-.0146 (.0124)	-.0533 (.0111)
D95	-.0677 (.0128)	-.0671 (.0119)
D96	-.1874 (.0133)	-.0111 (.0148)
D97	-.2702 (.0131)	-.0031 (.0145)
Nonsmoker	-.4596 (.0098)	-.1573 (.0079)
Male	.1867 (.0095)	.1035 (.0095)
Rated	.6140 (.0201)	.3365 (.0141)
R^{length}	1.4532 (.3612)	.8046 (.0494)
Participating	-.0001 (.0103)	-.0312 (.0092)
Own agent	.0955 (.0249)	.4629 (.0173)
Others	Amount, length, marital status	Amount, interest rate, marital status
Dummies	Age, state, occupation	Age, state, occupation
R^2	.837	.764
Observations	10,812	29,917

NOTE.—The dependent variable is the log of the annual premium per \$1,000 of face value of insurance. Variables are defined in the text. In addition to the coefficients listed, both regressions include the log of the real face value, the real face value, and the real face value squared; dummies if the face amount was censored at the maximum reported value; and dummies for marital status as well as age, state, and

TABLE 3
RESULTS BY CATEGORY: TERM PRICES

SAMPLE	STATE		OCCUPATION		AGE	
	Calif., Wash., Va.	Ala., La., Ky., Ark.	High Skill	Low Skill	<30	>45
	(1)	(2)	(3)	(4)	(5)	(6)
D93	.0801 (.0395)	.1580 (.0555)	.0439 (.0229)	.0697 (.0372)	.0857 (.0264)	.1239 (.0321)
D94	.0262 (.0377)	-.0399 (.0454)	-.0221 (.0215)	-.0195 (.0359)	-.0426 (.0257)	.0379 (.0325)
D95	-.0605 (.0354)	-.0788 (.0589)	-.1029 (.0220)	-.0171 (.0331)	-.1127 (.0290)	-.0095 (.0338)
D96	-.1932 (.0377)	-.0920 (.0503)	-.2030 (.0227)	-.1484 (.0384)	-.2530 (.0276)	-.0996 (.0328)
D97	-.3203 (.0411)	-.1254 (.0526)	-.3311 (.0218)	-.2293 (.0413)	-.3496 (.0260)	-.1411 (.0344)
R^2	.839	.828	.811	.866	.741	.820
Observations	1,451	623	3,347	1,297	2,248	205

NOTE.—The dependent variable is the log of the annual premium per \$1,000 of face value of insurance. All the regressions concern term life policies. The sample is restricted to the group listed at the top of the column. Variables are defined in the text. In addition to the coefficients listed, all the regressions include 20 variables and age, state, and occupation dummies. They are the same as those in table 1. Standard errors are in parentheses.

were about 1 percent lower in 1994 but almost 19 percent lower in 1996

Interpretation: Table 3 is a simple heterogeneity check. It does not yet use the continuous Internet-use variable, but it shows that the groups with faster Internet adoption had larger price declines.

Economic message: The timing and demographic pattern are consistent with Internet search lowering markups in the groups most likely to use it.

5.5 Table 4: Log real price as a function of Internet usage

Read: The Internet-use coefficient is negative across age-state, age-occupation, age-income, and occupation-state specifications. The age-state coefficient is about -0.5109 , implying that a 10 percentage point increase in Internet use lowers prices by roughly 5.1 percent in that specification.

Interpretation: This is the central regression table. The year effects shrink substantially after controlling for Internet use, meaning Internet diffusion explains a large share of the term-life price decline.

Economic message: The table provides the main evidence that reduced search costs increased competition and lowered transaction prices.

5.6 Table 5: Further controls

Read: Column 1 uses the insurance-research measure and obtains a large negative effect. Column 2 allows nonlinearity. Column 3 controls for group-level mortality. Column 4 estimates the effect for whole life. Column 6 checks early years before comparison sites. Column 7 examines policy riders.

Interpretation: The table addresses alternative explanations. The effect remains negative with mortality controls, does not appear for whole life, does not appear before comparison sites existed, and does not appear to be driven by riders.

TABLE 4
LOG REAL PRICE AS A FUNCTION OF INTERNET USAGE

Type	Age × State (1)	Age × Occupation (2)	Age × Income (3)	Occupation × State (4)
% use Internet	-.5109 (.1189)	-.2269 (.0955)	-.3454 (.1078)	-.1819 (.0860)
D93	.0606 (.0143)	.0600 (.0110)	.0118 (.0170)	.0605 (.0133)
D94	-.0100 (.0149)	-.0142 (.0160)	-.0301 (.0135)	-.0142 (.0121)
D95	-.0681 (.0130)	-.0669 (.0146)	-.0394 (.0151)	-.0672 (.0129)
D96	-.0515 (.0333)	-.1240 (.0289)	-.0955 (.0341)	-.1409 (.0269)
D97	-.0663 (.0499)	-.1757 (.0403)	-.1401 (.0454)	-.2005 (.0379)
R ²	.838	.837	.829	.838
Observations	10,812	10,812	8,676	10,806

NOTE.—The dependent variable is the log of the premium per \$1,000 of face value of insurance. All the regressions concern term life policies. The variable % use Internet is the share of the group that had Internet access in the given year. Variables are defined in the text. In addition to the coefficients listed, all the regressions include 20 variables and age, state, and occupation dummies. They are the same as those in table 1. Standard errors are in parentheses.

	Research (1)	Nonlinear (2)	Mortality (3)	Whole Life (4)	Interactions (5)	Early Years (6)	Pr(Rider) (7)
% research	-2.5360 (.5701)	-3.9660 (1.0120)	-2.5700 (.5689)	-3.876 (.5871)	-3.1570 (.4831)	-2.4360 (.3922)	.2230 (.4108)
(% research) ²		14.2280 (8.5030)					
% research (use from 1992-95)						.6110 (.7014)	
ln(mortality)			.1072 (.0392)				
D93	.0600 (.0161)	.0604 (.0145)	.0571 (.0144)	-.0597 (.0142)	.0499 (.0146)	.0587 (.0148)	-.0508 (.0040)
D94	-.0143 (.0128)	-.0143 (.0128)	-.0186 (.0129)	-.0335 (.0117)	-.0122 (.0117)	-.0188 (.0136)	-.0318 (.0028)
D95	-.0677 (.0129)	-.0677 (.0129)	-.0671 (.0131)	-.0671 (.0126)	-.0646 (.0132)	-.0765 (.0171)	-.0272 (.0060)
D96	-.1184 (.0161)	-.0909 (.0228)	-.1156 (.0159)	-.0215 (.0181)	-.1007 (.0200)	-.1176 (.0162)	-.0600 (.0120)
D97	-.1625 (.0214)	-.1351 (.0292)	-.1548 (.0216)	-.0177 (.0212)	-.1415 (.0263)	-.1620 (.0216)	-.0801 (.0170)
R ²	.838	.830	.829	.764	.803	.838	.082
Observations	10,812	10,812	10,812	29,917	10,812	10,812	11,499

NOTE.—The dependent variable is the log of the annual premium per \$1,000 of face value of insurance in all the columns except col. 7. Col. 4 concerns whole life policies, whereas all other columns concern term life policies. The variables are defined in the text. The dependent variable in col. 7 is the binary variable of whether the policy has a rider. Each regression also includes 20 variables and age, state, and occupation dummies. Standard errors are in parentheses. The % research variable in the years before the competition sites existed is the online usage in the year times the share of the group having researched insurance online in 1998. The results were quite similar when just % online was used instead.

Table 5: Further controls.

Economic message: The robustness pattern strengthens the interpretation that the mechanism is online comparison shopping, not general technological change or changing actuarial fundamentals.

5.7 Table 6: Price dispersion and Figure 2: Predicted standard deviation of price residuals

Read: Table 6 regresses within-group residual price dispersion on insurance research and its square and cube. The linear term is positive, the squared term is negative, and the cubic term is positive. Figure 2 shows the implied relationship: dispersion rises at low online research shares and then falls after about 5 percent.

Interpretation: The non-monotonic pattern matches search theory when the initially informed share is close to zero. Giving a small group of consumers full information can increase dispersion because firms split between high-price and low-price strategies. Once many consumers become informed, high prices become harder to sustain.

Economic message: The Internet made the market more competitive, but not by immediately eliminating dispersion. The transition itself created a temporary increase in price dispersion.

TABLE 6
PRICE DISPERSION

	STANDARD DEVIATION	
	(1)	(2)
% research	3.871 (.971)	3.477 (.981)
(% research) ²	-68.503 (29.503)	-50.555 (30.017)
(% research) ³	307.002 (203.984)	187.001 (205.370)
Constant	.264 (.005)	...
Dummies	None	Age, state, occupation
R ²	.028	.086
Observations	1,248	1,391

NOTE.—The dependent variable is the standard deviation of residuals from the price regression in col. 1 of table 2. Standard errors are in parentheses.

net has led to a change in the composition of policy type

(a) Table 6: Price dispersion.

34

JOURNAL OF POLITICAL ECONOMY

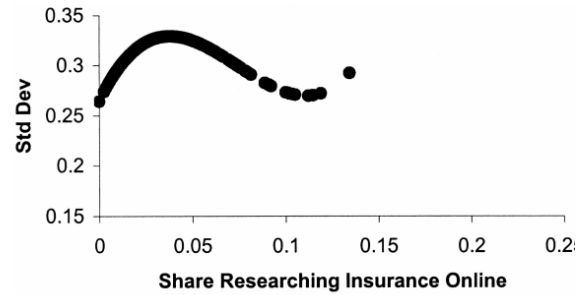


FIG. 2.—Predicted standard deviation of price residuals by age-state-year. Sources' calculations.

(b) Figure 2: Predicted standard deviation of price residuals.

6 Interpretive synthesis

- **Mechanism.** Internet comparison sites reduce search costs, making it easier to identify low-premium insurers.
- **Average-price effect.** Lower search costs reduce markups and lower transaction prices, especially for demographic groups that use the Internet more.
- **Product-specific effect.** Term life is affected because it is comparable online; whole life is not because it is heterogeneous and not covered by the comparison sites.
- **Timing effect.** Internet use before comparison websites existed does not explain price declines, which supports the comparison-site mechanism.
- **Dispersion effect.** Early adoption can increase dispersion; broader adoption eventually reduces dispersion.
- **Welfare effect.** The estimated 8–15 percent price reduction implies large consumer gains in a major financial product market.

7 Short summary

- The paper studies term life insurance from 1992–1997 using LIMRA policy-level data.
- It matches policy transactions to Forrester measures of Internet adoption and online insurance research by demographic group.
- Internet adoption predicts lower term life premiums after comparison websites appear.
- The estimated price reduction from 1995–1997 is about 8–15 percent.
- Whole life insurance, which was not covered by comparison sites, does not show the same price decline.
- Internet use before the sites existed does not explain price declines.
- Mortality controls and rider tests do not eliminate the effect.
- Price dispersion first rises and then falls as online research spreads.
- The paper contributes one of the earliest direct empirical estimates of how Internet search changes offline market competition.

8 Conclusion

8.1 Core conclusion

Brown and Goolsbee show that the Internet made the term life insurance market more competitive by reducing consumer search costs. As Internet comparison tools spread, term life prices fell significantly, while whole life prices did not show the same pattern.

8.2 Economic intuition

When search is costly, firms can sustain high prices because many consumers do not compare alternatives. Internet comparison sites increased the fraction of informed consumers. That shifted demand toward lower-priced insurers and pressured sellers to lower prices.

8.3 Incremental contribution in one sentence

The paper's distinctive contribution is to show, using transaction-level data and demographic Internet-adoption variation, that Internet search tools affected prices and dispersion in an offline financial product market.

8.4 Limitations

The authors cannot observe insurer identities, so they cannot control for firm fixed effects or fully rule out unobserved quality changes. They also observe transactions rather than the complete offer distribution, so they cannot fully separate firm price cuts from consumer switching toward cheaper firms.

8.5 Final takeaway

The Internet's effect on markets is not limited to online sales. By reducing search costs, online comparison tools can discipline offline firms, lower prices, change dispersion, and generate substantial consumer surplus in traditional financial markets.